Capstone Project – Economics Analysis

“What goes with higher income? Education, healthcare, labor and renewable energy?”

Education, healthcare, labor participation and renewable energy compared with GDP per capita. Six countries, 2000-2023.

USA | Australia | Canada | Russia | China | India

One question, five objectives, six countries

Problem Statement: "How can education, healthcare, labor participation, and renewable energy be leveraged to increase GDP per capita, while understanding the global economic, social, and environmental patterns that drive differences in development across countries?"

6	countries	**24**	years, 2000-2023
144	country-year rows	**34**	indicators

1. **Education**
   Literacy rate and education spending
2. **Healthcare**
   Healthcare spending and doctors per patient
3. **Labor participation**
   Participation and unemployment rates
4. **Renewable energy**
   Renewable share of energy
5. **Development level**
   HDI (Human Development Index) and poverty rate

Data and method: one dataset, two ways of looking

Data and cleaning

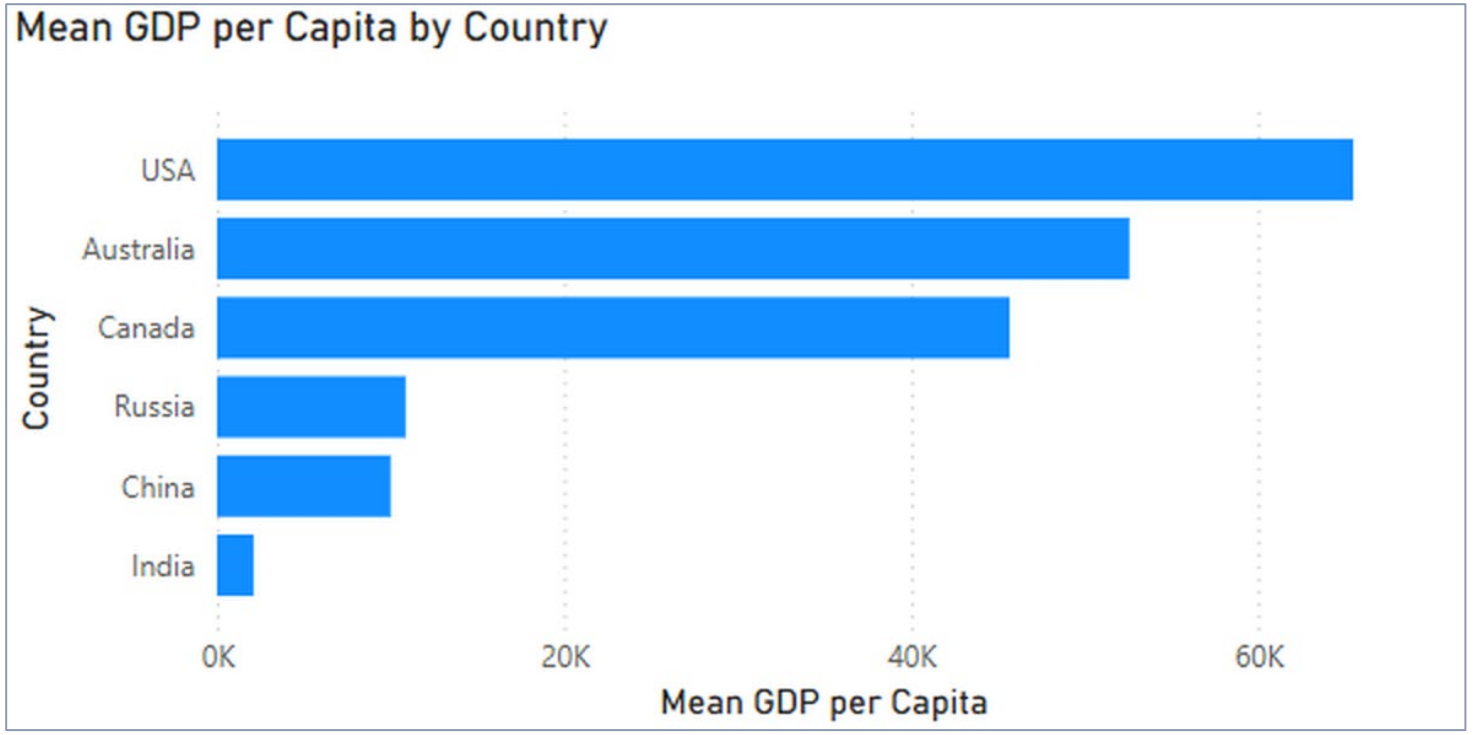
- Global Economic Data file: 144 rows (6 countries x 24 years) and 36 columns
- No missing values; 42 literacy values above 100% capped at 100 and 6 gender equality values above 1 capped at 1
- Countries span three income tiers: USA, Australia and Canada; Russia and China; India

How we analysed it

1. Pooled correlation: each lever against GDP per capita across 144 country-years

2. Linear regression with country as a category, 80/20 train/test split

Income differs enormously: the USA earns about 31 times India



Source: Power BI dashboard on Global Economic Data, 2000-2023. Mean GDP per capita over 24 years.

USD 46,000-65,500

USA, Australia and Canada

About USD 10,000-11,000

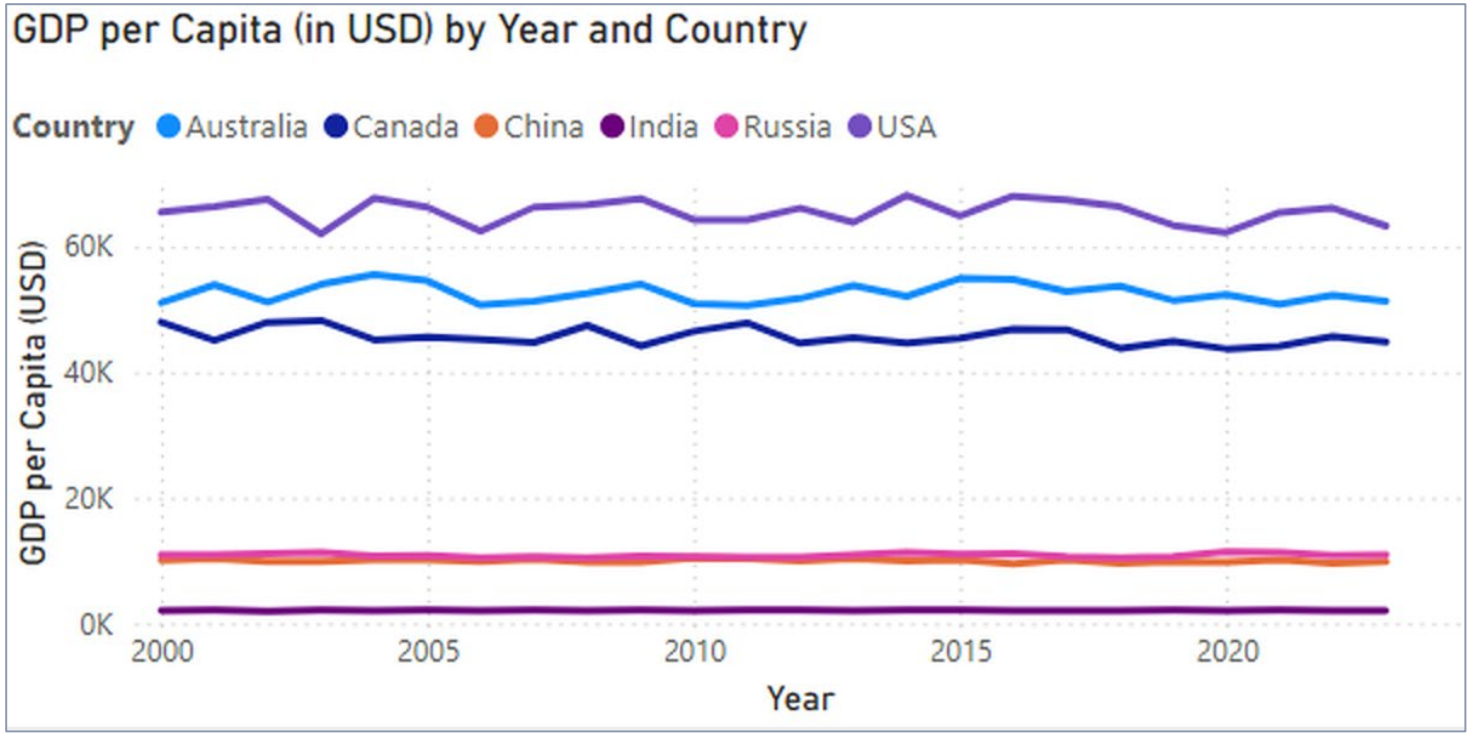
Russia and China

About USD 2,100

India

The question
What separates these three groups? The rest of this deck looks at the levers one by one.

Each country stays in its own income band for all 24 years



Source: Power BI dashboard on Global Economic Data, 2000-2023. GDP per capita by year.

About 10%

gap between each country's lowest and highest year, relative to its average

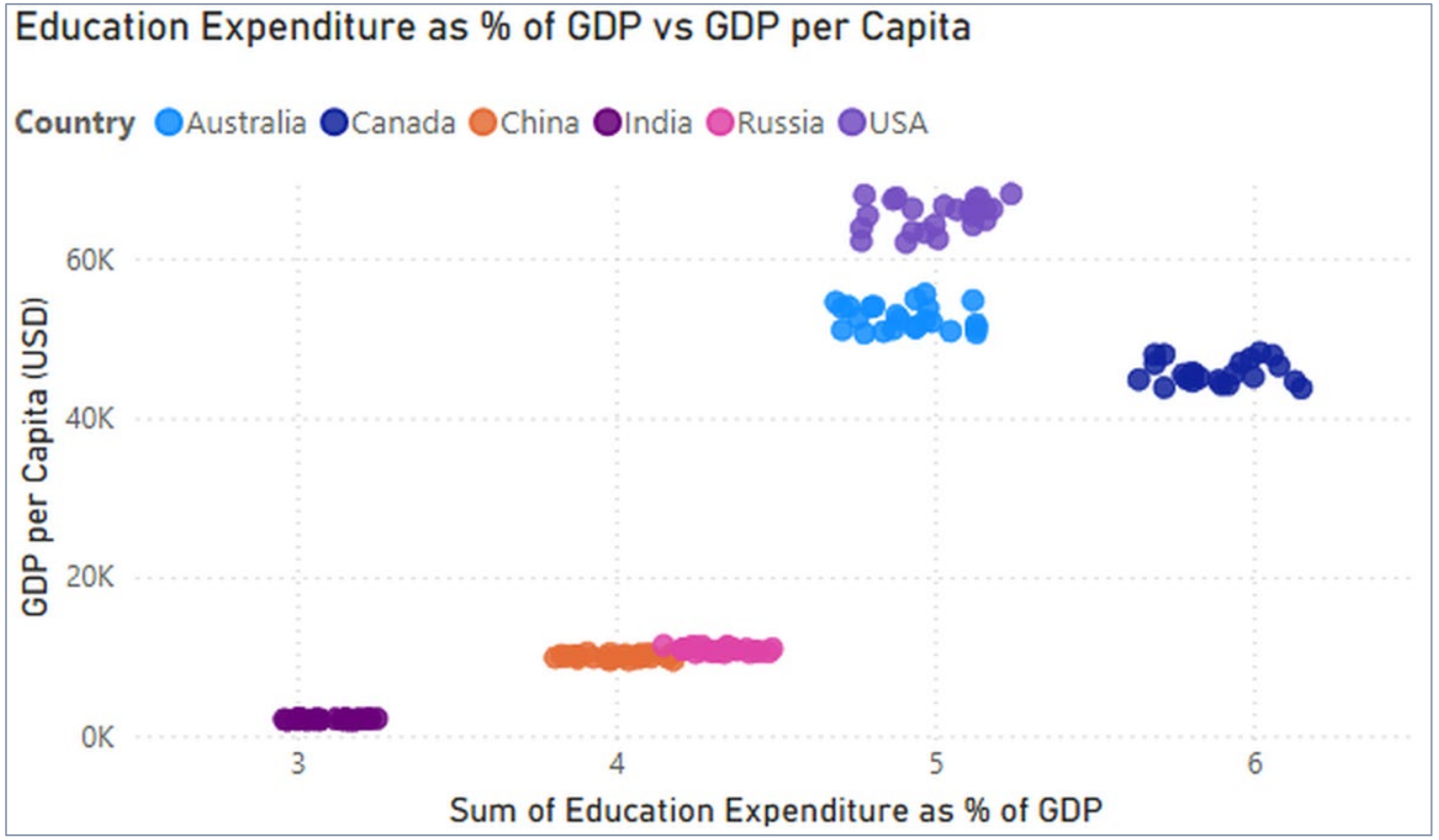
No overlap

between the income ranges of the three groups across all 24 years

What it means

Income differences between countries are far larger than changes over time, so the patterns that follow describe differences between countries.

Objective 1: Countries that spend more on education tend to be richer



Source: Power BI dashboard on Global Economic Data, 2000-2023. One dot per country-year.

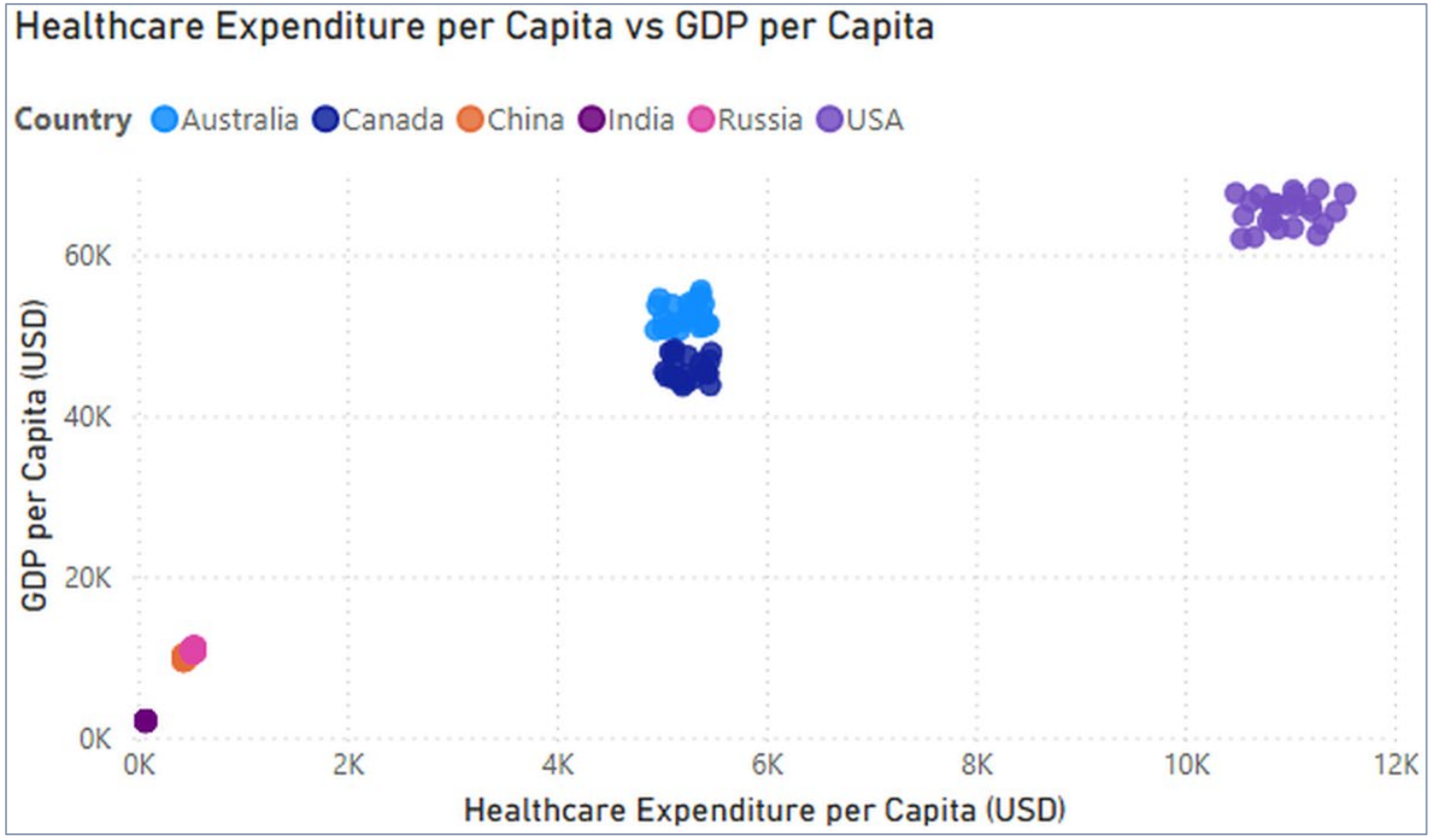
r = 0.79

education spending and GDP per capita, all six countries

- Strong link across countries: higher spending, higher income
- Canada spends the most (about 5.9%) but earns less than Australia and the USA
- Literacy is a weaker link (r = 0.56): five of six countries sit at 96-99%

Watch out
As a share/percentage of GDP, the same percentage is far more money per person in a rich country.

Objective 2: Healthcare spending is the closest marker of income



Source: Power BI dashboard on Global Economic Data, 2000-2023. One dot per country-year.

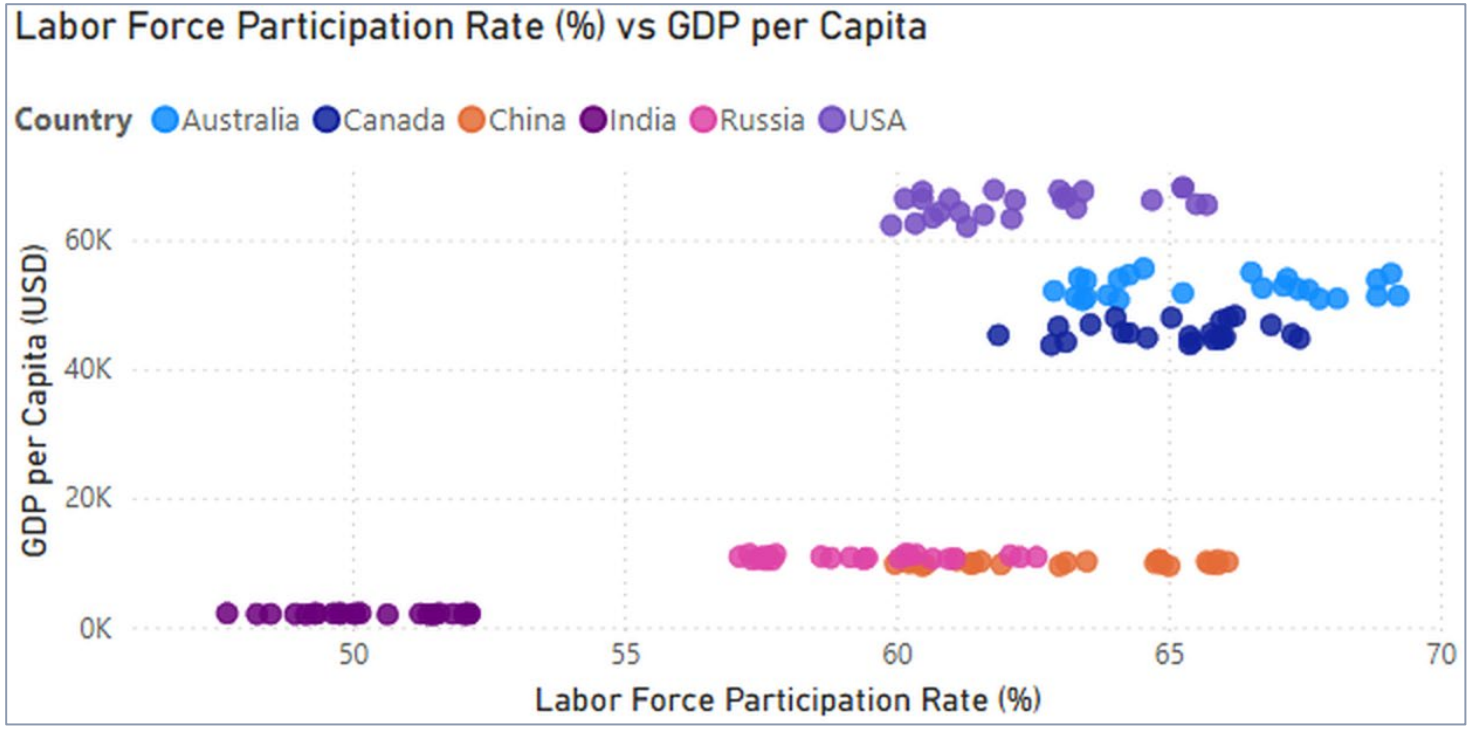
r = 0.95

healthcare spending per person and GDP per capita, all six countries

- Spending per person runs from about USD 73 in India to USD 10,960 in the USA
- Strongest link of any lever tested
- Doctors per patient is a weaker link (r = 0.40)

Watch out
Richer countries can afford to spend more, so the link may run both ways.

Objective 3: Labor participation shows only a moderate link with income



Source: Power BI dashboard on Global Economic Data, 2000-2023. One dot per country-year.

r = 0.64

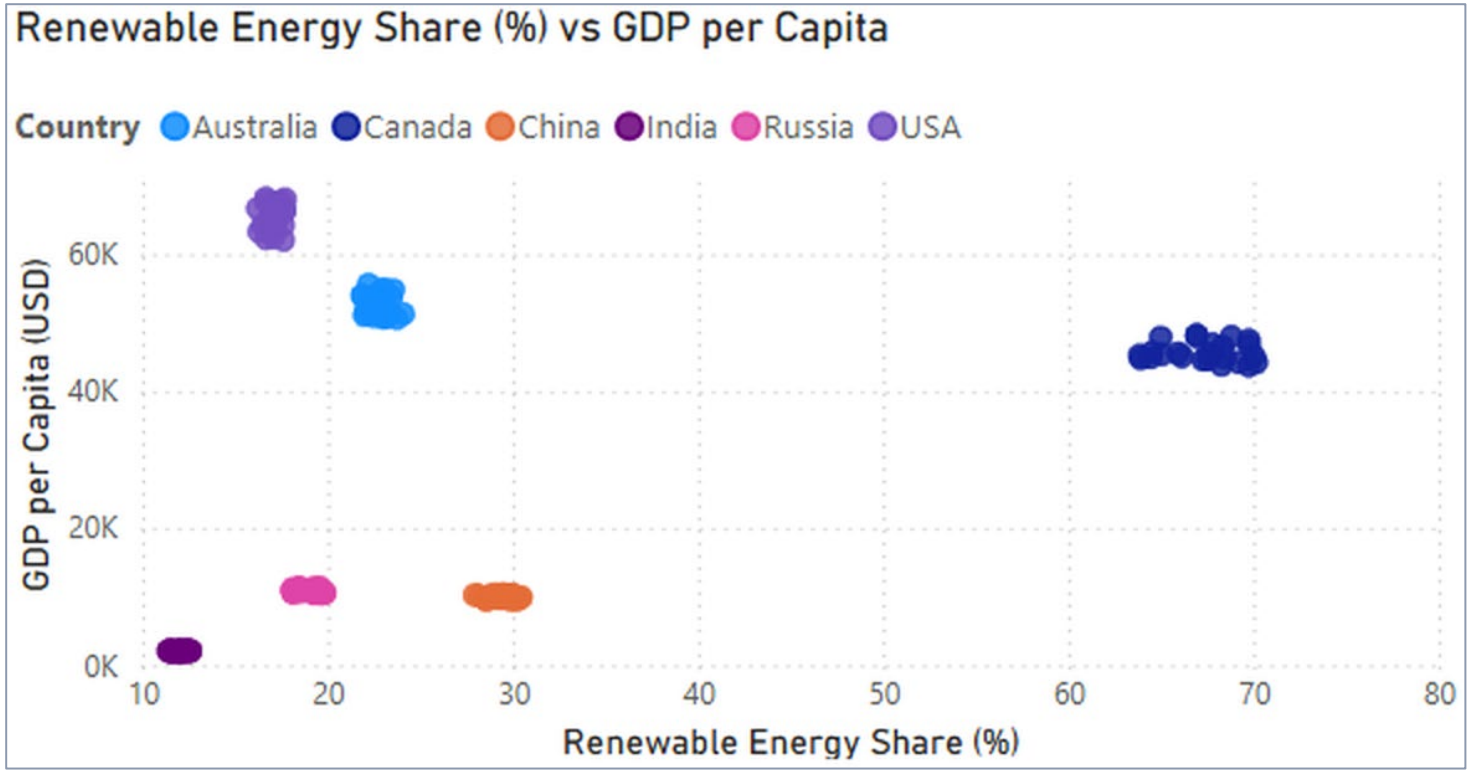
labor participation and GDP per capita, all six countries

- Most countries have similar participation (about 57-69%), yet incomes range from about USD 10,000 to USD 65,500
- China and the USA have similar participation at very different incomes
- Unemployment is a weaker link (r = -0.37)

How to read this

r = 0.64 is mostly driven by one country with low participation and low income. The other five show no clear pattern.

Objective 4: Renewable energy share shows no clear link with income



Source: Power BI dashboard on Global Economic Data, 2000-2023. One dot per country-year.

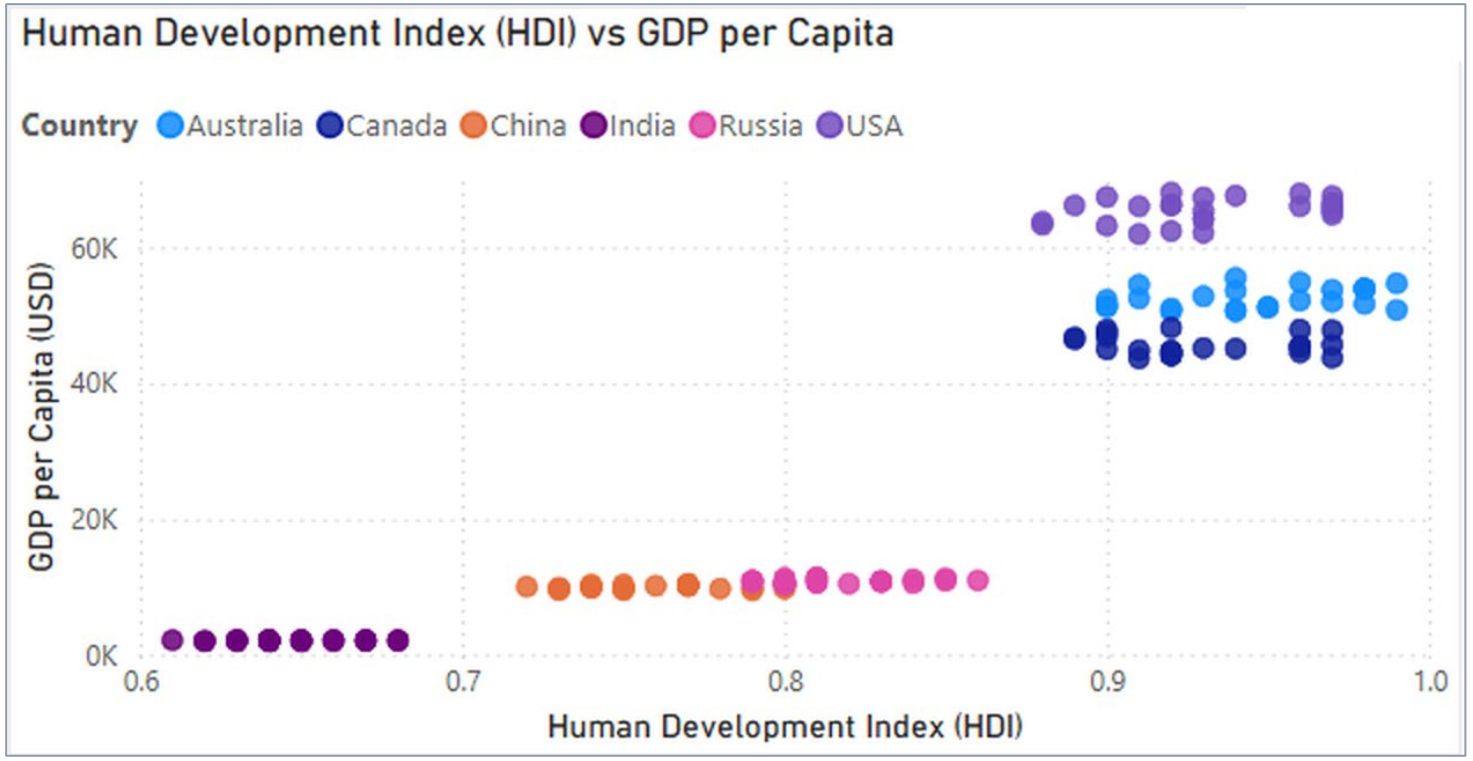
r = 0.26

renewable energy share and GDP per capita, all six countries

- Canada (about 67% renewable) earns less than the USA (17%) and Australia (23%)
- India has the lowest share and income; the other five show no order
- Renewable energy share doesn't predict income, either up or down

Watch out
Six countries is a small sample, so read this as no sign of a link, not proof of none.

Objective 5: HDI tracks income closely; poverty rate does not



Source: Power BI dashboard on Global Economic Data, 2000-2023. One dot per country-year.

r = 0.88

HDI and GDP per capita, all six countries

- HDI ranks the countries almost exactly as income does
- A strong link is expected, because HDI includes income
- Poverty rate is weak (r = -0.26); China is lowest (about 2%) at USD 10,000

What goes with higher income? Healthcare and education spending

Strong link

- Healthcare spending per person (r = 0.95)
- Education spending as a share of GDP (r = 0.79)

The highest-spending countries are the highest-income ones.

Moderate to no link

- Labor participation (r = 0.64)
- Literacy rate (r = 0.56)
- Doctors per patient (r = 0.40)
- Renewable energy (r = 0.26)
- Poverty rate (r = -0.26)

Mixed or weak across six countries; none as strong as healthcare and education.

Over time

- Each country stays in its own income band for 24 years
- Within countries, every lever has r between -0.04 and 0.17

The links describe differences between countries, not changes over time.

These are associations, not proven causes: richer countries can afford to spend more. ie. just because two variables are correlated doesn't mean one causes the other

eg. on hot days, more people buy ice cream, and more people go swimming. More swimming means more drownings. So ice cream sales and drownings rise together, but ice cream doesn't cause drowning

A simple model predicts income well

Linear regression predicting GDP per capita from eight levers plus country. HDI is left out because income is part of it. Random 80/20 split, stratified by country.

1. **Inputs**
   Eight levers, with country turned into 0/1 columns
2. **Result**
   Test R^2 0.997 against train 0.998: no overfitting gap
3. **Read with care**
   A high R^2 is not proof of cause: the model predicts income, it does not show what raises it

0.997	R^2 on test rows	**1,237**	USD error (RMSE) on test rows
115	training rows	**29**	test rows

Limits of the evidence and next steps

Limits

- Only six countries, so results depend heavily on which countries are included
- Associations, not proven causes: the link may run both ways
- HDI includes income, so it is used for description only
- Poverty rate and doctor-to-patient ratio are not defined in the data dictionary. ie. poverty line could be the percentage of people living below an international line or each country's national line. Doctor-to-patient ratio ie. per 1000 patients? per 10000 patients?
- Literacy values above 100% and Gender Equality values above 1 were capped and flagged

Next steps

1. Repeat the analysis with many more countries, using official data
2. Test education spending per person, not only as a share of GDP
3. Confirm how poverty rate and doctor-to-patient ratio are defined